

MENOMONIE, WI, USA

WMO: 720327

Lat: 44.892N

Lon: 91.868W

Elev: 273

StdP: 98.09

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04995

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.5	-22.0	-28.9	0.3	-23.8	-26.9	0.3	-21.2	10.7	-8.0	9.3	-9.0	2.4	270	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.0	22.0	30.0	21.4	28.0	20.2	24.1	29.0	23.0	27.6	22.0	26.5	5.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.8	26.7	21.5	16.7	25.5	20.7	15.9	24.7	73.8	29.3	69.4	27.3	65.8	26.7	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.3	8.4	7.5		-28.1	34.2	3.2	2.4	-30.4	35.9	-32.2	37.3	-34.0	38.6	-36.3	40.3	
				-28.3	25.8	3.1	1.1	-30.6	26.6	-32.4	27.2	-34.1	27.8	-36.4	28.6	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-9.0	-7.4	0.2	7.5	14.4	19.5	22.0	20.4	16.5	9.0	1.4	-6.2		
	DBStd	12.10	7.05	6.61	6.85	5.45	4.87	3.35	2.99	3.17	4.31	4.96	5.59	6.24		
	HDD10.0	2359	588	488	312	110	14	0	0	0	4	78	263	502		
	HDD18.3	4287	847	721	563	325	140	26	5	15	85	292	509	760		
	CDD10.0	1423	0	0	8	35	151	285	371	323	198	48	4	0		
	CDD18.3	309	0	0	0	1	18	61	117	80	29	3	0	0		
	CDH23.3	2755	0	0	4	20	212	532	1062	646	253	28	0	0		
	CDH26.7	779	0	0	0	2	62	138	349	168	56	3	0	0		
(14) Wind	WSAvg	3.2	3.3	3.3	3.3	3.9	3.5	3.1	2.6	2.5	2.9	3.3	3.4	3.2		
(15) Precipitation	PrecAvg	854	28	27	53	77	99	134	102	107	89	61	47	31		
	PrecMax	1047	67	56	132	161	177	228	232	225	226	146	184	61		
	PrecMin	680	5	9	10	20	46	60	26	11	13	16	0	8		
	PrecStd	109	17	13	28	37	41	45	42	59	50	40	42	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	10.6	22.5	26.2	32.1	32.5	35.0	32.6	31.1	26.6	19.5	8.2		
		MCWB	3.5	7.8	15.1	15.0	19.5	22.6	22.6	22.2	22.0	18.0	13.1	5.1		
	2%	DB	3.7	5.9	17.2	22.3	27.9	30.1	32.3	30.2	28.0	23.0	15.1	5.5		
		MCWB	1.7	3.3	11.8	12.9	17.8	20.8	22.7	21.8	20.4	16.8	10.6	3.8		
	5%	DB	2.2	3.0	12.9	19.7	25.8	27.8	30.1	28.3	26.4	20.7	12.2	2.9		
		MCWB	1.0	1.0	8.7	11.4	17.0	19.5	21.8	20.7	19.3	15.1	8.1	1.7		
	10%	DB	0.7	1.3	9.1	17.1	22.8	26.4	28.3	27.1	24.2	17.2	9.7	1.8		
		MCWB	-0.2	-0.1	6.0	9.9	15.0	18.8	20.8	20.0	18.8	12.5	6.9	0.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	7.0	16.2	16.2	22.6	24.9	26.0	24.6	23.7	21.5	14.2	7.3		
		MCDB	6.5	10.3	22.5	22.6	29.2	30.0	30.8	29.4	28.4	23.6	16.8	7.5		
	2%	WB	2.0	3.6	13.0	14.4	20.4	22.7	24.6	23.4	22.2	17.9	11.3	4.0		
		MCDB	3.3	5.7	15.0	20.1	25.2	27.3	29.7	27.8	26.0	21.0	15.2	5.2		
	5%	WB	0.9	1.4	8.9	12.5	18.7	21.4	23.2	22.4	20.8	15.4	8.7	2.1		
		MCDB	1.9	2.9	13.3	17.5	22.8	26.0	28.1	26.6	24.3	19.2	11.6	3.1		
	10%	WB	-0.3	0.0	6.2	10.7	16.9	20.2	22.1	21.5	19.6	13.1	6.7	0.9		
		MCDB	0.4	1.0	9.0	16.0	21.1	24.7	26.9	25.3	23.4	16.5	9.3	1.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.3	9.6	10.8	11.5	10.8	11.0	11.0	11.3	10.1	8.2	7.3		
		MCDBR	9.1	9.8	13.8	16.1	15.1	12.9	12.7	12.6	13.7	14.7	12.9	8.4		
	5% WB	MCWBR	7.7	7.6	8.5	8.5	6.9	5.8	5.2	5.6	6.7	8.2	8.4	7.0		
		MCWBR	8.7	9.1	12.7	14.0	12.1	11.1	11.1	10.6	11.0	12.0	11.7	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.285	0.318	0.358	0.393	0.394	0.382	0.391	0.362	0.322	0.289	0.264		
		taud	2.394	2.306	2.355	2.342	2.286	2.319	2.368	2.354	2.414	2.510	2.508	2.434		
	Ebn at Noon		875	902	916	903	877	875	881	859	858	855	831	829		
		Edh at Noon	72	96	106	118	129	125	118	115	100	79	66	64		
(44) All-Sky Solar Radiation	RadAvg		1.72	2.81	3.89	4.57	5.32	5.91	6.19	5.19	4.06	2.58	1.63	1.20		
	RadStd		0.18	0.27	0.38	0.47	0.42	0.46	0.24	0.35	0.27	0.33	0.15	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page